

Assignment 4

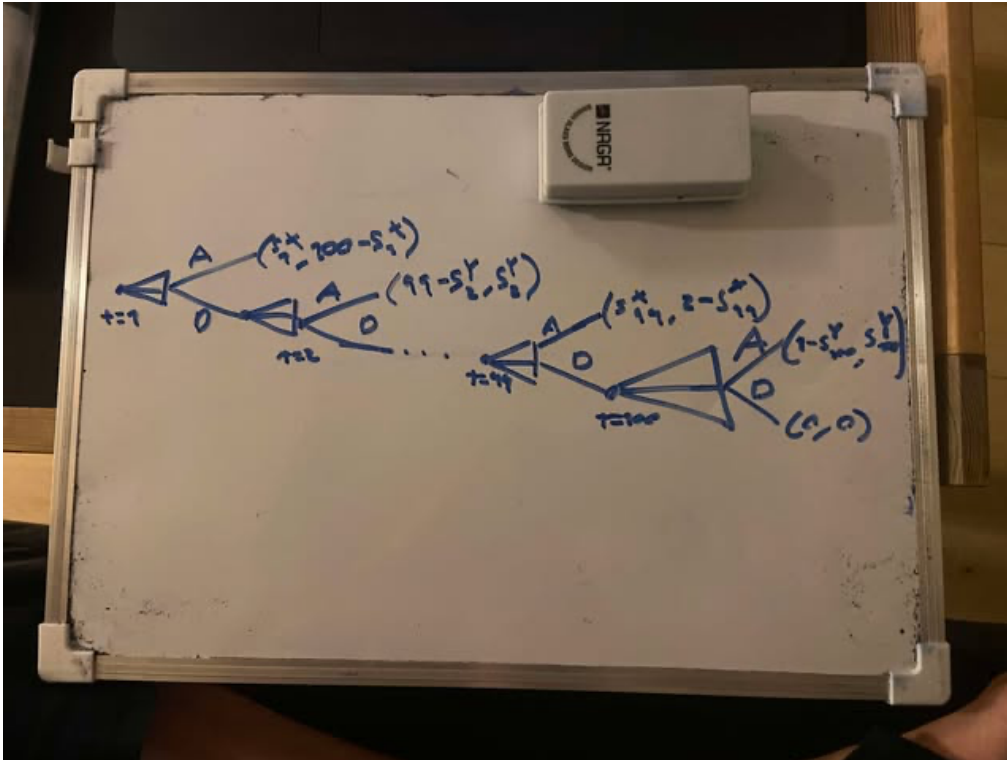
In round k an offer S is played by X if k is even or 0 and by Y if k is uneven

$S_k \in [0, 101 - k]$ being the payoff that player X gets and player Y getting $101 - k - S$

We solve using backwards induction in round 100 there is no more to be shared thus both players get 0 if we reach round 100

Here is the drawing

(notice i've swapped the notation because I found it confusing)



We reason subgame nash equilibria

In round 100 if $S \geq 0$ then X accepts thus $S_{99} = 0$ where Y gets 1

thus in round 99 Y accepts if he gets more than ≥ 1 giving us $S_{98} = 1$

we can write it a little more compact

$$100: (s_{100}^X, s_{100}^Y) = (0, 1)$$

$$99: (s_{99}^X, s_{99}^Y) = (1, 1)$$

$$98: (s_{98}^X, s_{98}^Y) = (1, 2)$$

These are all unique subgame EQ's

we note that a player will accept the amount of turns he has remaining leaving the rest for himself

This gives us the only subgame perfect EQ strategies

SPEQ: {offer counterparty his # of remaining turns and take the rest; -||-}

Ending in emidiate resolution

$$1: (s_1^X, s_1^Y) = (50, 50)$$